WORKED SOLUTION

STEP I 2002, Question 12

Probability and Statistics • Geometric Probability

STEP I 2002, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q12

The key idea

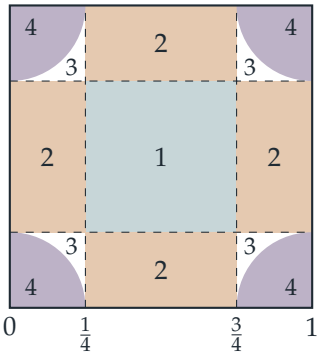
The grid repeats itself, so "placing the hoof at random" just means the centre of the hoof is uniformly distributed over one unit square. How many squares the hoof touches depends only on how close the centre is to the edges and corners of that square, so each answer corresponds to a region of the square, and its probability is the area of that region. Once we have the four probabilities, the rest is comparing them, and their mean, with what Harry actually does.

Solution

The distribution of Harry's answers

Let the centre C of the hoof lie in the unit square $0 \leq x \leq 1$, $0 \leq y \leq 1$. If Harry places his hoof at random, then by the periodicity of the grid C is uniformly distributed over this square, so the probability that C lies in a region is the region's area. Let N be the number of squares partly covered.

The hoof has radius $\frac{1}{4}$, which is less than $\frac{1}{2}$, so it can cross at most one vertical grid line and at most one horizontal one. It crosses a vertical line exactly when $x < \frac{1}{4}$ or $x > \frac{3}{4}$, and a horizontal line exactly when $y < \frac{1}{4}$ or $y > \frac{3}{4}$. (Positions exactly at distance $\frac{1}{4}$ from a line have zero area, so we ignore them.)



The events "crosses a vertical line" and "crosses a horizontal line" each have probability $\frac{1}{2}$, and they are independent because x and y are independent uniform coordinates. So:

- $N = 1$ when the hoof crosses no line: C is in the central square of side $\frac{1}{2}$, so $\mathrm{P}(N = 1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.
- $N = 2$ when it crosses exactly one line: $\mathrm{P}(N = 2) = 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$.
- When it crosses one line of each kind, C is in one of the four corner squares of side $\frac{1}{4}$, with total probability $\frac{1}{4}$. The two lines meet at a corner of the grid, and the hoof covers part of all four squares round that corner if and only if it covers the corner point itself.

Ask yourself In a corner square, why does the answer depend on the distance to the grid corner?

Take the corner square nearest the origin, with $0 < x < \frac{1}{4}$ and $0 < y < \frac{1}{4}$. The hoof certainly reaches the square containing C, the square to its left and the square below it. The diagonal square $x < 0$, $y < 0$ has nearest point the origin, at distance $\sqrt{x^2 + y^2}$ from C, so the hoof enters it exactly when $\sqrt{x^2 + y^2} < \frac{1}{4}$. So $N = 4$ when C is within $\frac{1}{4}$ of a grid corner, and $N = 3$ otherwise.

The points of the unit square within $\frac{1}{4}$ of one of its corners form four quarter-discs of radius $\frac{1}{4}$, of total area $\pi \left(\frac{1}{4}\right)^2 = \frac{\pi}{16}$. Hence

$$\mathrm{P}(N = 1) = \tfrac{1}{4}, \qquad \mathrm{P}(N = 2) = \tfrac{1}{2}, \qquad \mathrm{P}(N = 3) = \tfrac{1}{4} - \tfrac{\pi}{16}, \qquad \mathrm{P}(N = 4) = \tfrac{\pi}{16}.$$

Is the evidence so far consistent with a random horse?

When the correct answer is $4 \div 4 = 1$, a random horse is right with probability $\mathrm{P}(N = 1) = \frac{1}{4}$. When the correct answer is $2 + 2 = 4$, it is right with probability $\mathrm{P}(N = 4) = \frac{\pi}{16} \approx 0.196$. These are exactly the success rates observed, so yes, the evidence is entirely consistent with my suspicion that Harry is just placing his hoof at random.

The expected answer, and the new horse

If Harry places his hoof at random, his answer doesn't depend on the question, so the expected value of his answers is just $\mathrm{E}(N)$ (the equal frequency of the true answers plays no part):

$$\begin{aligned}\mathrm{E}(N) &= 1\cdot\tfrac{1}{4} + 2\cdot\tfrac{1}{2} + 3\left(\tfrac{1}{4} - \tfrac{\pi}{16}\right) + 4\cdot\tfrac{\pi}{16}\\ &= \tfrac{1}{4} + 1 + \tfrac{3}{4} + \tfrac{\pi}{16}\\ &= 2 + \tfrac{\pi}{16} \approx 2.20.\end{aligned}$$

Compare the observed average of 2 with two hypotheses.

- *Harry calculates correctly.* The true answers are $1, 2, 3, 4$ equally often, so the average should be $\frac{5}{2}$. An average of 2 is well below this, so Harry is certainly getting many answers wrong. In fact each wrong answer is at most 3 below the truth, and the average shortfall is $\frac{1}{2}$, so at least $\frac{1}{6}$ of his answers must be wrong.
- *Harry is random.* The average should be close to $2 + \frac{\pi}{16} \approx 2.20$. A single answer has variance $\mathrm{E}(N^2) - \mathrm{E}(N)^2 = \frac{9}{2} + \frac{7\pi}{16} - \left(2 + \frac{\pi}{16}\right)^2 \approx 1.05$, so over n problems the average has standard deviation about $1.02/\sqrt{n}$. With "many" problems (say $n = 400$, standard deviation about 0.05) an average of 2 is about four standard deviations too low, which is very unlikely by chance.

So over many problems, an average of 2 is too low even for a horse placing its hoof at random: Harry seems to favour small answers, as if he presses lightly or aims for the middle of a square. Either way, he is not doing the calculations, and he does worse than a random horse would. Yes: it's time for a new horse (though perhaps one that can do arithmetic).

Checks

- The four probabilities add up: $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} - \frac{\pi}{16} + \frac{\pi}{16} = 1$, and $\mathrm{P}(N = 3) = \frac{1}{4} - \frac{\pi}{16} \approx 0.054$ is positive, as it must be.
- A simulation of 400 000 random hoof placements, counting the squares within distance $\frac{1}{4}$ of the centre, gave relative frequencies 0.2495, 0.4990, 0.0547, 0.1968 and mean 2.199, against $2 + \frac{\pi}{16} \approx 2.196$.
- Sanity check: with a tiny hoof the answer would almost always be 1, and the mean would tend to 1. A larger hoof gives larger answers, and ours sits in between.

Common slips

- Assuming the hoof covers four squares whenever it crosses two grid lines. It also has to reach the corner where they meet, which is what makes π appear.
- Treating the average of 2 as evidence of randomness because it is "below 2.5".

Randomness predicts about 2.2, so the comparison has to be made with that value, taking account of the spread of the average.

Beyond STEP

This is *geometric probability*: when a point is uniform over a region, probabilities are areas, and the same idea gives Buffon's needle, whose answer famously also involves π. The final part is a first taste of hypothesis testing, using the central limit theorem to judge whether an observed mean is plausible under a model. Grimmett and Welsh, *Probability: An Introduction*, covers both continuous uniform distributions and the limit theorems behind this kind of argument.

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